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Investigating Web size metrics for early Web cost estimation

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Abstract

This paper's aim is to bring light to this issue by identifying size metrics and cost drivers for early Web cost estimation based on current practices of several Web Companies worldwide. This is achieved using two surveys and a case study. The first survey (S1) used a search engine to obtain Web project quote forms employed by Web companies worldwide to provide initial quotes on Web development projects. The 133 Web project quote forms gathered data on size metrics, cost factors, contingency and possibly profit metrics. These metrics were organised into categories and ranked. Results indicated that the two most common size metrics used for Web cost estimation were "total number of Web pages" (70%) and "which features/functionality to be provided by the application" (66%).

The results of S1 were then validated by a mature Web company that has more than 12 years of experience in Web development and a portfolio of more than 50 Web applications. The analysis was conducted using an interview.

Finally, once the case study was finished, a second validation was conducted using a survey (S2) involving local New Zealand Web companies. The results of both validations were used to prepare Web project data entry forms to gather data on Web projects worldwide. After gathering data on 67 real Web projects worldwide, multivariate regression applied to the data confirmed that the number of Web pages and features/functionality provided by the application to be developed were the two most influential effort predictors.
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1. Introduction

The World Wide Web (Web) has become the best known example of a hypermedia system. Numerous organisations world-wide are developing thousands of commercial and/or educational Web applications. Recently, growth of the Web as a delivery environment has given birth to a new research field—Web engineering, so applying engineering principles to develop quality Web applications (Pressman, 2000). A variety of technological solutions are available for Web developers to facilitate the delivery of quality Web applications and bring them quickly to market, with typical durations ranging from three to six months (Reifer, 2000; Pressman, 2000). However, there are no standardised development techniques or large data sets of historical data on Web development projects.

For Web development, cost is difficult to estimate because:

- There is no standard to sizing Web applications. Each can be created using diverse technologies such as several forms of Java (Java, Servlets, Enterprise Java Beans, Applets, and Java Server Pages), HTML, JavaScript, XML, XSL, and so on. Attempts have been made to apply Function points principles to sizing Web applications (Rollo, 2000; Mendes et al., 2002), however Mendes et al. did not find any

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improvement in estimation accuracy using Function points-based size metrics, when compared to other size metrics (e.g. number of Web pages, number of images).

- Web development processes differ substantially from traditional approaches (Reifer, 2002; Baskerville et al., 2003). A recent survey of Web development practices by Barry and Lang (2001) showed that practitioners are not using development processes proposed in the literature (e.g. UML). They often develop their own in-house methods, which often include prototyping.
- Web project's primary goal is to bring quality applications to market as quickly as possible, varying from a few weeks to six months (Pressman, 2000; Reifer, 2002; Offutt, 2002).
- People involved in Web development are represented by less experienced programmers, users as developers, graphic designers and new hires straight from university (Reifer, 2002; Standing, 2002).
- Processes employed are in general ad hoc (Standing, 2002), although some organisations are starting to look into the use of agile methods (Ambler, 2002).

In terms of the types of Web applications we consider for our research, we include two categories of applications which broadly represent the applications delivered using the Web: Web hypermedia applications and Web software applications (Christodoulou et al., 2000). A Web hypermedia application is a non-conventional application characterised by the authoring of information using nodes (chunks of information), links (relations between nodes), anchors, access structures (for navigation) and its delivery over the Web. Technologies commonly used for developing such applications are HTML, JavaScript and multimedia. In addition, typical developers are writers, artists and organisations who wish to publish information on the Web and/or CD-ROM without the need to use programming languages such as Java. These applications have great potential in areas such as software engineering (Fielding and Taylor, 2000), literature (Tosca, 1999), education (Michau et al., 2001), and training (Ranwez et al., 2000).

Conversely, a Web software application represents conventional software applications that depend on the Web or use the Web's infrastructure for execution. Typical applications include legacy information systems such as databases, booking systems, knowledge bases etc. Many e-commerce applications fall into this category. Typically they employ development technologies (e.g., DCOM, ActiveX etc.), database systems, and development solutions (e.g. J2EE). Typical developers are young programmers fresh from a Computer Science or Software Engineering degree, managed by more senior staff.

The motivation for documenting both types of applications is to help readers understand Section 2 and to avoid conflict of terminology with other papers.

We use Web application to refer to both Web hypermedia and software applications.

This work investigates metrics that can be used for Web cost estimation at an early stage of a Web development project. One example would be that of estimating effort at the bidding stage, where application requirements are not yet fully known. Similarly to Kitchenham et al. (2003) we consider a project price quote to have three components: estimated cost, profit and contingency. We are interested in metrics used to estimate cost (effort) at an early stage, and therefore leaving aside profit and contingency.

The rest of the paper is organised as follows:

Section 2 provides an overview of Web size metrics aimed at Web cost estimation. This overview is given such that we can compare it to our findings (discussed in Section 4), and also to provide detailed descriptions that can help other researchers who are conducting systematic literature reviews in the field (Kitchenham et al., 2004).

Section 3 presents two surveys, a case study, and how the results of these three investigations were used to inform a research project named Tukutuku. The objective of the first survey (S1) was to identify size metrics and cost factors used at an early stage of a Web development project for Web cost estimation. We used data collected from 133 Web project price quote forms from companies worldwide. The sample population was that of Web companies that use Web forms to give price quotes on Web development projects. The gathered metrics were organised into six categories: three (Web application static metrics, Web application dynamic metrics, Cost drivers) consisted of candidates for size metrics and cost drivers for early effort estimation; one (Web project metrics) incorporated metrics we believe can have a bearing on the contingency and/or profit costs that are provided in a price quote (Kitchenham et al., 2003) and to help set priorities and decide on the skills necessary/available to the project; one (Web company metrics) included metrics that characterise a customer; one (Web interface style metrics) contains metrics related to an interface's style. The objective of the case study and second survey (S2) was to validate the results obtained in S1. Validation in the scope of our work represents identifying size metrics and cost factors important to be used in the Web cost estimation process early in the development life cycle. The first validation used a case study and consisted of interviewing a mature Web Company. The second validation employed a survey investigation (S2), where the sample population was that of Web companies in New Zealand.

Section 4 presents the results of a multivariate regression analysis using data from 67 industrial Web projects

from the Tukutuku database. It also compares the regression analysis results to those presented in Sections 2 and 3. The database used is part of the Tukutuku research project, which aims to gather data from Web companies to benchmark development productivity and to develop cost estimation models. The Web project data entry form employed in the Tukutuku project was prepared based on feedback from S1, S2 and the case study mentioned above. The two out of three effort predictors selected for the best fit regression model correspond very closely to the two highest ranked size metrics from S1—Web pages and features/functionality provided by an application.

Finally Section 5 gives our conclusions and comments on future work.

2. Web size metrics for Web cost estimation

To date few papers have proposed Web size metrics aimed at Web cost estimation (Cowderoy et al., 1998; Mendes et al., 1999; Cowderoy, 2000; Mendes and Mosley, 2000; Reifer, 2000; Rollo, 2000; Cleary, 2000; Mendes et al., 2001). Cowderoy (2000), Cowderoy et al., (1998), Reifer (2000) and Cleary (2000) have used industrial data sets of Web projects to justify their size metrics and to generate corresponding cost models, each collecting their data from just one Web company, possibly affecting the external validity of their results. Mendes et al. (2001) proposes size metrics for static and dynamic Web applications and Mendes and Mosley (2000) proposes size metrics for Web hypermedia applications. However the data sets employed in these studies are based on Web applications developed by students, which may also affect the external validity of their results.

Each of these papers is presented in the following subsections chronologically.

2.1. Size metrics by Cowderoy et al. (1998) and Cowderoy (2000) [1]

Cowderoy (1998, 2000) recommends several size metrics for cost estimation and risk assessment of Web application development projects. Metrics were organised by the Entities to which they apply.

Web application

Web pages (WP): the number of Web pages in an application.

Home pages (HP): the number of major entry points to the Web application.

Leaf nodes (LN): the number of leaf Web pages in an application, i.e., pages that have no siblings.

Hidden nodes (HN): the number of Web pages excluded from the main navigation buttons.

Depth (DE): the number of pages on the second level that have siblings.

Application Paragraph count (APC): the sum of Page paragraph count for all Web pages in an application.

Delivered images (DI): the number of unique images used by the Web application.

Audio files (AF): the number of unique audio files used in a Web application.

Application movies (AM): the sum of Page movies for all the Web pages in an application.

3d objects (3DO): the number of files including one or more 3D objects used in a Web application.

Virtual worlds (VW): the number of files including one or more virtual worlds used in a Web application.

External hyperlinks (EH): the number of unique URLs in the Web application.

Web page

Actions (AC): the number of independent actions provided from a Web page by use of Javascript, ActiveX and Java applets.

Page paragraph count (PPC): the number of paragraphs in a Web page.

Word count (WC): the number of words in a Web page.

Navigational structures (NS): the number of different structures in a Web page.

Page movies (PM): the number of movie files used in a Web page.

Interconnectivity (IN): the number of URLs that link to other pages in the same application.

Media

Image size (IS): computed as width × height.

Image composites (ICS): the number of layers from which the final image was created.

Language versions (LV): the number of versions of an image that must be produced to accommodate different languages or different cultural priorities.

Duration (DU): summed duration of all sequences within an audio file.

Audio sequences (AS): number of sequences within the audio file.

Imported images (IMI): the number of separate graphics images imported into an interactive audio file.

Program

Lines of source code (LOC): the number of lines of code in a program/script.

McCabe ciclomatic complexity (MCC) (Fenton and Pfleeger, 1997): the structural complexity of a program/script.

[1] A detailed description of these measures is given in http://www.xvc63.dial.pipex.com/my-complexity/index.shtml.

2.2. Size measures by Mendes et al. (1999, 2001), Mendes and Mosley (2000)

Mendes et al. (1999, 2001), Mendes and Mosley (2000) propose size metrics to be used to predict authoring effort for hypermedia applications and then for Web applications. All metrics are presented organised by Entities to which they apply.

Hypermedia application
Hyperdocument size (HS): the number of documents that the hypermedia application has. Documents are considered here to be either HTML files or any kind of file that is defined as a document in the hypermedia systems used in the evaluations.
Connectivity (CON): the number of links that the hypermedia application has. These links can be internal or external. Dynamically generated links are excluded.
Compactness (Botafogo et al., 1992) (COM): measures how inter-connected the nodes are.
S*tratum* (Botafogo et al., 1992) (STR): measures to what degree the hypermedia application is organised for directed reading.
Link generality (LG): measures if the link applies to a single instance, for example point-to-point links, or whether it applies (or can be applied) to multiple instances.

Later Mendes and Mosley (2000) proposed a new set of size metrics, all targeting at Web applications:

Page count (PAC): the number of HTML or SHTML files used in a Web application.
Media count (MEC): the number of unique media files used in a Web application.
Program count (POC): the number of CGI scripts, JavaScript files, Java applets used in a Web application.
Total page allocation (TPA): the total space allocated for all HTML or SHTML pages (Mbytes) used in a Web application.
Total media allocation (TMA): total space allocated for all media files (Mbytes) used in a Web application.
Total code length (TCL): total number of lines of code for all programs used in a Web application.
Reused media count (RMC): the number of reused or modified media files used in a Web application.
Reused program count (RPC): the number of reused or modified programs used in a Web application.
Total reused media allocation (TRM): total space allocated for all reused media files used in a Web application (Mbytes).
Total reused code length (TRC): total number of lines of code for all programs reused by a Web application.
Code comment length (CCL): total number of comment lines in all programs in a Web application.
Reused code length (RCL): total number of reused lines of code in all programs in a Web application.
Reused comment length (ROL): total number of reused comment lines in all programs in a Web application.
Total page complexity (TPC): the average number of different types of media used in the Web application, excluding text.
Connectivity (CON): measures the total number of internal links, not including dynamically generated links.
Connectivity density (COD): computed as *Connectivity* divided by *page coun*t.
Cyclomatic complexity (Fenton and Pfleeger, 1997) (CCO): computed as *Connectivity page coun*t + 2.

This list was revised to include also bottom-up metrics (Mendes et al., 2001):

Web page
Page allocation (PAL): measures the allocated space of a HTML or SHTML file (Kbytes).
Page complexity (PCO): the number of different types of media used on a page, not including text.
Graphic complexity (GRC): the number of graphics media used in a page.
Audio complexity (AUC): the number of audio media used in a page.
Video complexity (VIC): the number of video media used in a page.
Animation complexity (ANC): the number of animations used in a page.
Scanned image complexity (SIC): the number of scanned images used in a page.
Page linking complexity (PLC): the number of links per page.

Media
Media duration (MED): the duration of audio, video, and animation (minutes).
Media allocation (MEA): the size of a media file (Kbytes).

Program
Program code length (POL): the number of lines of code in program.

2.3. Size metrics by Rollo (2000)

Rollo (2000) did not suggest any new size metric, however, as far as we know, he was the first researcher to investigate the issues of measuring the size of Web hypermedia and Web software applications, aiming at cost estimation, using several function point analysis methods. He measures the size of two applications in

both *IFPUG*, *MKII*, and *COSMIC-FFP* [2] methods. Rollo (2000) concludes that *COSMIC-FFP* proved to be the most flexible approach for counting the functional size of Web hypermedia and Web software applications and can be applied to any Web application.

2.4. Size metrics by Cleary (2000)

Cleary (2000) proposes size metrics for Web cost estimation dividing them into two types: size metrics for Web hypermedia applications and size metrics for Web software applications.

Web hypermedia application
Non-textual elements (NTE): the number of unique non-textual elements within an application.
Externally sourced elements (ESE): the number of externally sourced elements.
Customised infra-structure components (CIC): the number of customised infra-structure components.
Total Web points (TWP): the total size of a Web hypermedia application in Web points by adding:

- number of Web pages of "Low" complexity multiplied by the value for "Low" Web points; with
- number of Web pages of "Medium" complexity multiplied by the value for "Medium" Web points; with
- number of Web pages of "High" complexity multiplied by the value for "High" Web points.

Web software application
Function points (FPS): the functionality of a Web software application. Does not specify any particular method.

Web page
Non-textual elements page (NTP): the number of non-textual elements within a Web page.
Words Page (WOP): measures the number of words in a Web page.
Web points (WPP): the length of a Web page. Scale points are "Low", "Medium" and "High". Each scale point is attributed a number of Web points, previously calibrated to a specific Web projects data set.
Number of links into a Web page (NIL): the number of incoming links; can be internal or external links.
Number of links out of a Web page (NOL): the number of outgoing links; can be internal or external links.
Web page complexity (WPC): the complexity of a Web page based upon its *number of words*, and combined *number of incoming* and *outgoing links*, plus the *number of non-textual elements*. The scale points are "Low", "Medium" and "High". Value ranges are provided for each scale point, for number of words and combination of *incoming links + outgoing links + non-textual elements*. These values have been calibrated based on a specific Web projects data set.

2.5. Size measures by Reifer (2000)

Reifer (2000) proposes a size metric called Web Objects, which measures the number of Web Objects. Size is measured using an adaptation of Halstead's equation for volume, tuned for Web applications. The equation is as follows:

$$V = N\log_2(n) = (N_1 + N_2)\log_2(n_1 + n_2) \quad (1)$$

where N is the number of total occurrences of operands and operators; n is the number of distinct operands and operators; N_1 is the total occurrences of operand estimator; N_2 is the total occurrences of operator estimators; n_1 is the number of unique operands estimator; n_2 is the number of unique operators estimators; and V is the volume of work involved represented as Web Objects.

Operands are comprised of the following metrics:

Number of building blocks (NBB): number of components, e.g., ActiveX, DCOM, OLE.
Number of COTS (NCO): number of COTS components (including any wrapper code).
Number of multimedia files (NMM): number of multimedia files, except graphics files (text, video, sound etc.).
Number of object or application points (Cowderoy et al., 1998; Cowderoy, 2000) (NOA): the number of object or application points or others proposed (# server data tables, # client data tables etc.).
Number of lines (NLI): number of xml, sgml, html and query language lines (# lines including links to data attributes).
Number of Web components (NCM): number of applets, agents etc.
Number of graphics files (NGR): number of templates, images, pictures etc.
Number of scripts (NSC): number of scripts for visual language, audio, motion etc.

All size metrics presented in this Section were invariably related to implemented Web applications. Even when targeted at measuring functionality based on function point analysis, researchers only considered the final Web application, rather than requirements documentation generated using any existing Web development methods. This makes their usefulness as early effort predictors questionable. The surveys and case study presented in Section 3 aim to bring light to this issue, not

[2] COSMIC-FFP = COmmon Software Measurement International Consortium-Full Function Points.

only by identifying size metrics and cost drivers based on current practices of several Web companies worldwide, but also by comparing these identified metrics to those that have been proposed in the past, looking for possible convergence.

3. Identifying Web size metrics and cost factors for early Web cost estimation

3.1. First survey: using on-line Web project price quote forms

The purpose of this survey (S1) was to identify Web size metrics and factors used by Web companies to estimate effort for Web projects early on in the development cycle. The target population was that of Web companies that offer on-line Web project price quotes to customers. There was no need to contact Web companies directly, only to download their on-line Web project price quote forms from the Web.

The four research questions we were interested in answering were as follows:

Question 1. What are the size metrics and factors used as early effort estimators by Web companies?
Question 2. Which size metrics are used the most?
Question 3. Which size metrics are characteristic of Web hypermedia applications?
Question 4. Which size metrics are characteristic of Web software applications?

To obtain a sample of our population of interest, we used a Web search engine (www.google.com) to carry out a search, on the 7th of August 2002, using the sentence "quote Web development project". Google.com was chosen because at the time it was the fourth most popular search engine in the world, where numbers 1 and 3 both licence Google technology for their Web searches. About half of all Web searches in the world are performed with Google, and is translated into 86 languages. Our objective was to obtain Web addresses (URLs) for Web companies that had on-line price quotes for Web development projects where these quotes could be used to identify size metrics and cost factors. Two hundred and fifty nine Web addresses were retrieved. Of those, 38 did not contain any forms or useful description. Of the remaining 221 URLs, 88 presented on-line quotes that were too general to obtain any useful information. An example of a general Web form is given in Fig. 1.

3.1.1. Survey results

The data collected from 133 on-line quotes was organised into six categories:

- Web application static metrics.
- Web application dynamic metrics.
- Cost Drivers.
- Web project metrics.
- Web company metrics.
- Web interface style metrics.

Web application static metrics correspond to attributes that have been commonly used to size Web hypermedia applications (e.g., number of Web pages, number of graphics etc.). Web application dynamic metrics refer to any features or functionality mentioned (e.g., database integration, on-line secure order form etc.), which is taken as an abstract measure of functionality and therefore size (Fenton and Pfleeger, 1997). Cost drivers correspond to attributes that may influence effort but do not characterise application size. Web project metrics incorporated metrics we believed may have a bearing on the contingency and/or profit costs that are provided in a price quote (Kitchenham et al., 2003) (e.g., project budget). Web company metrics are attributes that characterise a Web company (e.g., target audience), and Web interface style metrics are attributes that characterise the final appearance (style) of the Web application (e.g., background colour, style etc.). The full list of metrics per category is given in Table 1. Table 1 also shows the percentage and corresponding number of Web companies that have included that metric in their quote form. Two metrics stood out, *total number of Web pages*

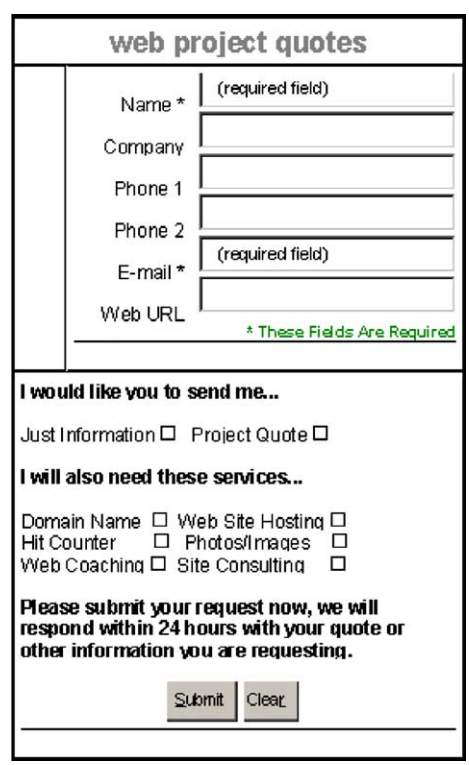

Fig. 1. Example of a general Web form with a very simple quote.

Table 1
Metrics organised by category and ranking

Category	Metrics	%	# companies
Web application static metrics	1. Total number of Web pages	70	92
	2. Total number images provided by customer	31	41
	3. Total number of text pages provided by customer	30	39
	4. Total number of photos provided by customer	30	40
	5. Total number of products to sell	28	37
	6. Total number of flash animations	27	35
	7. Total number of photos from a library	23	30
	8. Total number of images from a library	22	29
	9. Total number of audio clips	20	27
	10. Total number of video clips	20	26
	11. Total number of photos to scan	15	20
	12. Total number of images to scan	14	19
	13. Total number of gif animations	14	18
	14. Total number of photos to process/create	11	15
	15. Total number of text pages to type	11	13
	16. Total number of images to process/create	9	12
	17. Total number of icons/buttons	8	10
	18. Total number of text pages to scan (OCR)	8	10
	19. Total number existing components to add	4	5
	20. Total number of PDF/Word documents to download	2	3
	21. Total amount of Graphics per product	2	3
	22. Average length audio clips	2	2
	23. Average length video clips	2	2
	24. How much text necessary for each product	1	1
Web application dynamic metrics	Which features/functionality[a]	66	87
Cost Drivers	1. New application or enhancement	31	41
Web project metrics	1. Budget for the project	56	74
	2. Project estimated end date	35	46
	3. Web application type[b]	34	44
	4. Project estimated start date	32	42
	5. Will customer provide page mock-ups?	4	4
	6. Will the application be translated to different idioms?	3	4
	7. Will customer provide website map?	3	4
	8. What kind of Internet access most likely for customers?[c]	1	1
Web company metrics	1. Contact information (e-mail)	100	133
	2. Description of company's business	19	25
	3. Company's target audience[d]	18	24
Web interface style metrics	1. What is the preferred style for the site[e]	17	23
	2. Give three URLs for sites you like	17	23
	3. What type of colours for the site?[f]	13	17
	4. What type of colours for the background?[g]	5	7
	5. What type of colours for pages?[h]	5	7

[a] All have been considered as functionality/features for the sake of our ranking. Features/functionality are: Auction/Bid utility, Bulletin Boards, Discussion Forum/Newsgroups, Chat Rooms, E-postcard, Email-a-Friend, Mailing List for Site, Subscriptions, Vote System, Web-based e-mail, Database creation, Database integration, Other persistent storage integration (e.g. flat files), Credit Card Authorization, Member login, Online Secure Order Form, Password protected pages, Online Content Updating, Online feedback form, Shopping Cart Services, Invoices/Billing, Calendars/Event Calendar, Display current date, CSS (Style sheets) Charts, File upload/download, HR Handbooks, Job Accounting, Job Scheduling, Specialized Reports, Live data (stock quotes etc.), Performance reviews, Time Tracking, Traffic Statistics, Knowledge Base, Language Translation, Page counters, Personalisation, Search Engine, User Guest book, Visitor statistics, Cell Phone Optimisations, Palm PDA Web Clipping.

[b] Academic, Corporate, eCommerce, eTrading, Educational, Entertainment, Extranet, Family, Functional, Intranet, Media, Multimedia Presentation, News and Information, Non-Profit, Online community/forum, One Page Presence, Personal, Political, Portal (vertical or horizontal), Professional, Promotional, Search engine, Short-Term Promotion, Small Business, Support site/video chat, Virtual marketplace (B2B).

[c] T1 or better, DSL, IDSN, Dial Up (56K), Dial Up (33.6K).

[d] Business people and professionals, mainly women, mainly men, teenagers and twenty something, children, students, college or trade schools, anyone and everyone.

[e] Whimsical, bright and vivid, techno-futuristic, gritty, aggressive, mass-consumer oriented, really 'clean' looking, sports-oriented, fun and wacky, for women only, other.

[f] Primary colours, pastel colours.

[g] Solid colour background, image background.

[h] Light colour pages, dark colour pages.

and which *features/functionality*. Both can be taken as size metrics where the first is a typical *length size metric* and the second an abstract measure of *functionality* (Fenton and Pfleeger, 1997).

Seventy four Web companies also asked for the available Web project budget. We believe this metric can have a bearing on the contingency and/or profit costs that are provided in a price quote (Kitchenham et al., 2003). Project estimated end date, project estimated start date and application type also were important. We believe these help set priorities and perhaps decide on what skills are necessary and available to the project.

Answering Survey 1 research questions:

- What are the size metrics and factors used as early effort estimators by Web companies?

Size metrics identified in the survey are presented in Table 1, under *Web application static metrics* and *Web application dynamic metrics*. Those that have been ranked highest are possibly the most important, given that they were suggested by the greatest number of Web companies. For static metrics, Web companies require from the beginning how much information (text, photos, images), which is "ready to use", is provided by customers, how much information needs to be digitized or needs to be created from scratch. Number of pages was also important information. We believe this to be the case as most Web companies use number of pages as one of the core components for Web applications (e.g., 5 pages with text only, 5 pages with text, images etc.). Regarding dynamic metrics, Web companies listed features and functionality, rather than using a more sophisticated mechanism to measure functionality. This may suggest development practices where functional components are not well defined at the start of the project.

- Which size metrics are used the most?

According to the survey, the two metrics used the most are *total number of Web pages* and which *features/functionality*. Together they represent 70% and 66% respectively of those Web companies included in the survey. It is interesting to note that *total number of Web pages* represents a static size metric, characteristic of Web hypermedia applications and *which features/functionality* represents more closely an abstract measure of functionality, characteristic of Web software applications. We believe this is an attractive result as it seems to suggest a balance between static and dynamic characteristics which may be present in Web applications. In addition, *total number of Web pages* has also been the size metrics chosen by most literature described in Section 2. Conversely, the measurement of functionality, as suggested in the previous studies described in Section 2, is quite different from what the survey results suggest.

- Which size metrics are characteristic of Web hypermedia applications?

The category *Web application static metrics* characterise closely Web hypermedia applications.

- Which size metrics are characteristic of Web software applications?

The category *Web application dynamic metrics* characterise closely Web software applications.

The survey presented in this Section identified metrics related to a project's price quote. However, price is the result of three components (Kitchenham et al., 2003): estimated effort, contingency and profit. Therefore, since our objective was to identify only those metrics specifically targeted at effort estimation, we employed a case study and a second survey (S2) to identify the subset of metrics obtained in S1 directly related to effort estimation. The case study is presented in Section 3.2 and the second survey is presented in Section 3.3.

3.2. Case study: validating the results obtained for S1

The case study involved an experienced Web company contacted to validate, based on the ranking provided in S1, those metrics they consider important for early Web effort estimation. As mentioned previously, validation here represents identifying size metrics and cost factors important to be used in the Web effort estimation process early in the development life cycle.

This company was established in 1993 and has to date a portfolio of more than 50 Web applications. They document their development process and record size and effort metrics from past projects. Depending on the type of project, one out of two types of process models are used: prototyping or waterfall. The choice depends on the Company's familiarity with the application domain. The Company's effort estimation practices are based solely on expert opinion, where the average estimation accuracy for their Web projects, based on effort estimates obtained early in the life cycle, is 10%.

The types of services they provide are: Web application design, Web application development, Web application enhancement, e-commerce/shopping cart Development, Database Integration, Flash Animation Design, Graphic Design, Web application Maintenance, Training and Consulting.

The types of industry representative of their customers are Finance/Investment/Brokerage, Government, Insurance, Legal, Non-Profit, Telecommunications and Utilities.

This company is based in Rio de Janeiro, Brazil. Five people work in Web design and development, of which three are company directors, who, in addition to other tasks, also manage Web projects, prepare quotes for customers and develop applications.

One of the authors communicated over email with one of the directors, has worked in software development for more than 20 years and is experienced in management of large projects, conventional or Web-based. He was presented the results of S1 and asked to identify metrics from S1 that were actually employed by the company on their Web effort estimation process.

For *Web application static metrics* agreement was reached for most metrics (from number 1 to 18, see Table 1). More were added to the list, and some were re-arranged, as presented in Table 2.

For *Web application dynamic metrics* more features/functionality were added to the list and the director suggested that adding a complexity level to each feature/functionality would help discern more difficult implementations. More specifically, this Company groups functions/features within three groups: simple, complex and very complex. Each has an associated baseline, which represents a percentage to be added to estimated effort. The Company's baselines reflect average percentages based on past experience. These metrics were also confirmed as suitable for early Web effort estimation.

For *Web Project metrics*, numbers 1, 2, 3 and 4 (see Table 1) are employed by the Company for deciding on a project's contingency/risks and profit components (1, 2 and 4 for profit and 3 for risks/contingency). Therefore, these metrics are important for pricing, however not for effort estimation.

For *Web company metrics*, number 2 (see Table 1) is also taken into consideration for deciding on a project's profit component.

None of the *Web interface style metrics* were seen as relevant for effort estimation. These metrics, despite being related to an application's interface, do not include the interface's building blocks (e.g. graphics, text, audio, and video) which are directly related to interface design.

Jørgensen and Sjøberg (2001) found that the Web companies they have interviewed place a strong emphasis on interface design and this activity incorporates a large amount of the total effort spent in developing an application. The metrics we have obtained as a result of our first survey (S1), under the category Web application Static Metrics, and validated by the mature Web company corroborate their findings. Table 2 shows under the "graphics" group several size metrics that are directly related to interface design and the implementation of graphics to be added to a given Web application. Although none of the "graphics" metrics identified using S1 have been used by more than 31% of the Web forms we used, they were confirmed as important effort predictors by the mature Web company and by all Companies that participated in the second survey (S2), to be described in the next Section.

Table 2
Size metrics after first validation

Size metrics
Web pages
Total number of Web pages in the Web application
Number of Web pages created from scratch (new Web pages)
Number of Web pages provided by the customer
Number of outsourced Web pages
Total number of text pages (A4 size) that had to be typed: (Take each A4 Page to be 600 words)
Number of text pages provided by the customer in electronic format
Number of text pages provided by the customer that had to be scanned with OCR
Graphics
Total number of new images/photos created: (images also include icons and buttons)
Number of images/photos provided by the customer in electronic format
Number of images/photos provided by the customer that need to be scanned
Total number of images obtained from an image/photo library or outsourced
Total number of new animations (Flash/gif/3D etc.) created from scratch
Total number of animations (Flash/gif/3D etc.) obtained from a library or outsourced
Total number of new audio/video clips created
Total number of audio/video clips obtained from a audio/video library or outsourced
Total number of products to sell

3.3. Survey S2: Validating the results obtained for S1

The purpose of this second survey was also to validate the results that have been obtained from S1. We have also considered in this survey some of the results obtained from the case study, more specifically regarding Web application dynamic metrics.

The target population was that of Web companies in New Zealand that estimate effort for their Web projects. The survey instrument was a questionnaire with nine questions was prepared by one of the authors, and the method of gathering data was via interviews over the phone. A pilot study involving three Web companies from overseas was used to validate a previous version of this questionnaire. Based on their feedback it was decided that the questionnaire should not have too many questions in order to engage as many Web Companies as possible to participate. We have also discussed the possibility of using a Web-based on-line questionnaire instead, and emailing invitations to Web companies in New Zealand, followed by a telephone call. However, this choice was discarded since nowadays Companies in general often receive many, even hundreds of unsolicited emails, which we believe would reduce dramatically the chances of obtaining volunteers for our survey, even if we employed a follow-up telephone call.

Contact information was obtained for 100 Web companies in New Zealand, using the NZ Yellow pages and all companies were contacted directly over the phone. Thirty-two Companies (32%) accepted to be interviewed. Remaining companies declined the invitation on the basis of being busy, or not interested. The questionnaire took on average 15–20 min to answer.

We organised the Web application static size metrics into four categories (e.g. Web pages, Images/photos, animations and audio/video clips) and provided more general questions, such as the one presented below. This was done in order to have a simpler and smaller questionnaire.

Question 1. If you use application size to estimate the effort to develop a Web application, what aspects of size do you consider?

- The size of the application in number of images/photos.
- The size of the application in number of animations (Flash/gif/3D/2D).
- The size of the application in number of audio/video clips.
- The size of the application in number of Web pages.
- Other?

All the 32 companies use all four categories of size to estimate effort to develop a Web application.

In relation to the Web application dynamic metrics, rather than listing all functions/features we obtained from S1 and the case study, we chose to ask the following two questions:

Question 2. If you use the functions/features to be added to a Web application to estimate the effort to develop this application, how do you identify what functions/features to develop?

- Do you give clients a list of functions/features and ask them to pick the ones they want?
- Do you ask clients what they want without using a list or offering suggestions?
- Do you ask clients what they want and offer suggestions without using a list?
- Other?

Question 3. Do you differentiate between complex and simple functions/features to estimate effort? Yes No

For question number two, 19 companies (59%) provide a list of functions/features for clients to choose from. Thirteen companies (41%) ask clients what functions/features they want and offer suggestions without using a list.

For question number three, seven companies (22%) differentiate between complex and simple functions/features to estimate effort. However, the remaining 25 (78%) assume all functions/features to have the same complexity.

The remaining six questions used in our questionnaire are as follows:

Question 4. How many people work in your Company?

Question 5. Do you generally prepare a quote for a project or do you charge by time?

Question 6. Do you refine your effort estimate later on in the development?

Question 7. What factors do you use to estimate the effort for a project?

(a) Size of the application.
(b) Functions/features to be offered by the application.
(c) Experience of the developers with the development environment.
(d) Development tools used.
(e) Project Budget.
(f) Project estimated end date.
(g) Project estimated start date.
(h) Web application type.
(i) What is the preferred style for the site.
(j) What type of colours for the site.
(k) What type of colours for the background.
(l) What type of colours for pages.
(m) Customer will provide page mock-ups.
(n) Application will be translated to different idioms.
(o) Customer will provide website map.
(p) Internet access most likely for customers.
(q) Description of company's business.
(r) Company's target audience.
(s) Other(s)

Question 8. Do you use the same factors to estimate effort for new and enhancement projects?

Question 9. At what phase of a project does your Company prepare an initial effort estimate?

(a) Before requirements gathering phase.
(b) During requirements gathering phase.
(c) After requirements gathering phase.
(d) During design phase.
(e) After design phase.
(f) During implementation phase.
(g) Other(s):

Relative to question four, Fig. 2 shows a histogram for number of people per Company. The companies surveyed had from one person to 20 people, with median of 4.5 and mean of 5.44. Sixteen Companies (50%) have no more than four people, and 25 companies (78.1%) have no more than six people, suggesting that Web companies may fit the profile of small companies.

For question five, 30 Companies (94%) always charge by project and the other two (6%) charge by time spent.

For question six, 12 Companies (38%) do not refine their effort estimate, whereas the remaining 20 (62%) refine their estimates, although not often. This may indicate that at least the Web companies we surveyed

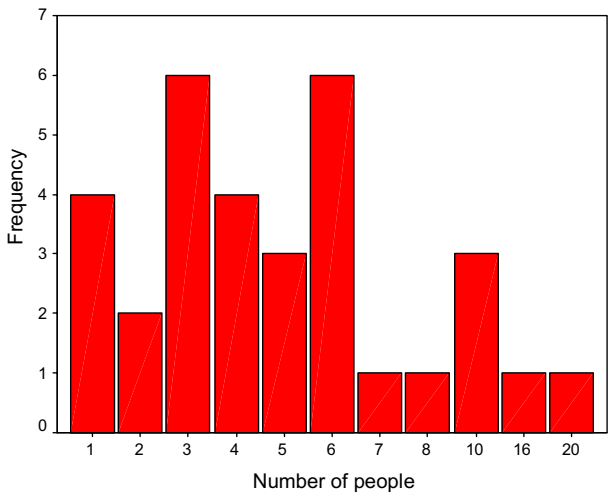

Fig. 2. Histogram of number of people per company.

use their initial effort estimate as their "final" effort estimate and adjust their work to fit their initial estimate. Our results corroborate those of Jørgensen and Sjøberg (2001).

For question seven, eight companies (25%) used size, functions/features, experience and tools. The remaining 24 companies (75%) all used size and functions/features. However only occasionally experience and tools were used as the people involved in Web development projects would remain unchanged and so would the tools employed. None of the Web project metrics, Web Company metrics or Web interface style metrics were selected as factors used for effort estimation. Eight Companies (25%) also suggested an extra factor they employ when preparing an effort estimate: how well known the requirements are. We believe that for early effort estimation, most notably for effort estimation to be used at the bidding stage (Kitchenham et al., 2003), this factor would have similar value throughout projects since early on in a project requirements are seldom well known.

For question eight, 13 Companies (41%) answered that they always consider the same factors for estimating effort for new or enhancement projects; nine Companies (28%) use different factors and 10 Companies (31%) sometimes employ the same factors for either new or enhancement projects and sometimes employ different factors. If we relate these answers to those of question seven, we can argue that size and functions/features are factors used by all Companies when estimating effort for either new or enhancement projects. What differs is the use of experience and/or tools.

For question nine, 10 Companies (32%) prepare an effort estimate during the requirements gathering phase and 20 Companies (62%) prepare an effort estimate during the Design phase. Two Companies (6%) did not answer this question since they do not have to provide customers with effort estimates, i.e., customers always pay for the Companies' actual effort.

The results of our second survey are summarised in Table 3.

3.4. Using feedback to inform the Tukutuku benchmarking project

The feedback obtained from Surveys 1 and 2 and the case study was used to prepare Web forms to gather data on Web projects worldwide. This data gathering initiative was called the Tukutuku benchmarking project. [3] The Tukutuku project aims to gather data on Web projects worldwide to be used to develop Web cost estimation models based on early effort predictors and to benchmark productivity across and within Web Companies.

While preparing the project data entry forms used in Tukutuku, careful consideration was given to differentiate more complex features/functions from less complex ones, as this was the current practice of some of the Web companies surveyed in S2 and also by the mature Web company from our case study. We had a detailed list of features/function obtained from S1 and the case study. Although some certainly seemed more complex than others we did not want to suggest or impose any complexity, leaving it for each Web company to decide. The solution was devised as follows:

- Companies contributing Web project data to Tukutuku would be asked to indicate (tick) all the features/functions that the application has.
- For each feature/function they would also indicate if it was a black box reuse, reuse with adaptation, or new development.
- They were also asked to indicate if a given feature/function employed high effort to be developed/adapted.
- To be familiar with what each Company understood by *high effort* we also asked them to indicate the effort in person hours that would be automatically representative of high effort to develop or adapt a feature/function.
- Companies would be able to provide features/functionality that we had not considered.

The Tukutuku benchmarking project started officially on October 2002. Invitation letters were e-mailed to Web companies in different countries and, in some cases (e.g. New Zealand, UK) with a follow up phone call a few days later, seeking participation. To date we have received 67 Web projects from 25 Web companies

[3] http://www.cs.auckland.ac.nz/tukutuku

Table 3
S2 metrics organised by category

Category	Metrics	%	# companies
Web application static metrics	1. Total number of Web pages	100	32
	2. Total number of images/photos (images also include icons and buttons)	100	32
	3. Total number of animations (Flash/gif/3D etc.)	100	32
	4. Total number of audio/video clips	100	32
Web application dynamic metrics	Which features/functionality	100	32
Cost Drivers	1. New application or enhancement	59	19
	2. Experience of development team	25+	8+
	3. Tools employed	25+	8+
Web project metrics	1. Budget for the project	0	0
	2. Project estimated end date	0	0
	3. Web application type	0	0
	4. Project estimated start date	0	0
	5. Will customer provide page mock-ups?	0	0
	6. Will the application be translated to different idioms?	0	0
	7. Will customer provide website map?	0	0
	8. What kind of Internet access most likely for customers?	0	0
Web Company metrics	1. Description of company's business	0	0
	2. Company's target audience	0	0
Web Interface Style metrics	1. What is the preferred style for the site	0	0
	2. What type of colours for the site?	0	0
	3. What type of colours for the background?	0	0
	4. What type of colours for pages?	0	0

in nine different countries. Twenty-seven projects come from two companies (13 and 14 respectively). Each Web project in the database provided 43 variables to characterise a Web application and its development process (see Table 4). We are aware that the data obtained is a result of a self-selected sample. However we believe the data in the Tukutuku database can be very useful as an indicator, provided one is aware of the limitations.

No automated measurement tools were used by the Web companies that volunteered data for the Tukutuku database. Therefore the accuracy of their data cannot be determined. In order to identify guesstimates from more accurate effort data, we asked companies how their effort data was collected (see Table 5).

Two companies used different data collection levels depending on the type of project (i.e. they used level 1 for some projects and levels 3 and 4 for other projects). Of the two companies that volunteered more than 10 projects, one used level 3 to record effort for all its 13 projects and the other used level 4 to record effort for all of its 14 projects. At least for 77.6% of Web projects in the Tukutuku database effort values were based on more than guesstimates. However, we are also aware that the use of timesheets does not guarantee 100% accuracy in the effort values recorded.

The data collected to date for the Tukutuku project has not followed rigorous quality assurance procedures to validate the data and the projects' application domains are mixed.

4. Using multivariate regression to identify early Web size metrics and cost factors

All the statistical analyses reported in this paper were accomplished using SPSS v.10.01.

The technique employed to build the cost model presented in this paper was forward stepwise regression.

The set of variables used for building the cost models is shown in Table 6. This is a subset of the Tukutuku data set since several variables had to be excluded if they were within the following constraints:

- Most instances of a variable were zero.
- The variable was categorical.
- The variable was related to another variable, in which case both could not be included in the same model. This was investigated using a Spearman's rank correlation analysis ($\alpha = 0.05$). The choice of a non-parametric correlation analysis is justified since most variables were not normally distributed.

Whenever variables were highly skewed they were transformed to a natural logarithmic scale to approximate a normal distribution (Maxwell, 2002). In addition, whenever a variable needed to be transformed but had zero values, the natural logarithmic transformation was applied to the variable's value after adding 1.

The cost model presented here was initially proposed and detailed in Mendes and Kitchenham (submitted for publication).

Table 4
Variables for the Tukutuku database

Variable name	Scale	Description
Country	Nominal	Country company belongs to
Established	Ordinal	Amount of time company has been established
Services	Nominal	Services Company provides
ClientInd	Nominal	Industries representative of clients
TypeProj	Nominal	Type of project
TypeApp	Nominal	Type of Web application developed
Languages	Nominal	Implementation languages used
nlang	Ratio	Number of different languages used
DocProc?	Nominal	Project followed defined and documented process
ProcImpr?	Nominal	Development team involved in a process improvement programme
Metrics?	Nominal	Development team part of a software metrics programme
devteam	Ratio	Size of development team
teamexp	Ratio	Average team experience with the development language(s) employed
Webpages	Ratio	Number of web pages
newWP	Ratio	Number of New Web pages
Wpcustom	Ratio	Web pages given by the customer
Wpout	Ratio	Web pages developed by third party
WpOwnCo	Ratio	Web pages reused from own company
txtTyped	Ratio	Number text pages typed (~600 words)
txtElec	Ratio	Number text pages electronic format
txtScan	Ratio	Number text pages scanned
imgNew	Ratio	Number new images
Img3rdP	Ratio	Number images developed by third party (not the customer)
imgScan	Ratio	Number images scanned
imgLib	Ratio	Number images reused from a library
imgOwnCo	Ratio	Number of images reused by own company
Animnew	Ratio	Number new animations
AnimLib	Ratio	Number animations reused from a library
AVNew	Ratio	Number new audio/video files
AVLib	Ratio	Number reused audio/video files
TotDiffPro	Ratio	Number ⟨⟩ products application offers
HEffDev	Ratio	Effort considered high to develop a single function/feature[a] by one person
HEffAdpt	Ratio	Effort considered high to adapt a single function/feature[b] by one person
hfots	Ratio	Number of reused high effort features/functions without adaptation
hfotsA	Ratio	Number of adapted high effort features/functions
hnew	Ratio	Number of new high effort features/functions
tothigh	Ratio	Total number high effort features/functions
fots	Ratio	Low effort features off-the-shelf
fotsa	Ratio	Low effort features off-the-shelf adapted
new	Ratio	Number new low effort features/functions
totnhigh	Ratio	Total number low effort features/functions
toteffor	Ratio	Total effort develop the Web application
accuracy	Nominal	Procedure used to record effort data

[a] This number is currently set to 15 h based on the collected data.
[b] This number is currently set to 4 h based on the collected data.

Table 5
What effort data was gathered

Effort data	Level	Number of projects and companies	
		# and % projects	# different companies
No timesheets	1	12 (17.9%)	8
Total hours worked each day or week	2	3 (4.5%)	3
Hours worked per project per day/week	3	24 (35.8%)	12
Hours worked per project task per day	4	28 (41.8%)	8

The model's stability was verified using two steps, which were (Maxwell, 2002):

Step 1. Use a residual plot showing residuals versus fitted values to investigate if the residuals are random and normally distributed.

Step 2. Calculate Cook's distance values for all projects to identify influential cases. Any projects with distances higher than $4/n$, where n represents the total number of projects, were considered to have high influence on the results. When there were projects with high influence the

Table 6
Variables used in the stepwise regression

Variable	Meaning
lntoteff	Natural log of the total effort to develop a Web application
nlang	Number of different languages used on the project
devteam	The number of people who worked on the project
teamexp	Average team experience with the development language(s) employed
lnnewwp	Natural log of (1 + number of new Web pages)
lnimgnew	Natural log of (1 + number of new images in the applications)
Lnimglib	Natural log of (1 + total number of images reused from a library)
lnimg3p	Natural log of (1 + total number of images developed by a third party)
hfotsa	Total number of adapted high effort functions
lntoth	Natural log of (1 + total number of high effort functions)
totnhigh	Total number of low effort functions

Natural log = Natural logarithm

stability of the model was tested by removing these projects, and observing the effect their removal had on the model.

The cost model is presented in Table 7. Its adjusted R^2 was 0.67.

The equation as read from the final model's output is

$$\ln(\text{toteffor}) = 2.154 + 0.435 \times \ln(\text{newWP} + 1) + 0.671 \times \ln(\text{tothigh} + 1) + 0.239 \times \text{devteam} \quad (2)$$

which, when transformed back to the raw data scale, gives the equation:

$$\text{toteffor} = 8.619 \times (\text{newWP} + 1)^{0.435} \times (\text{tothigh} + 1)^{0.671} \times e^{0.239 \times \text{devteam}} \quad (3)$$

When there is no reuse, Webpages is the same as newWP.

The model's prediction accuracy was checked by using the 'omit one project at a time' procedure (leave one out cross-validation). The rationale is to use different sets of projects to build and to validate a model. Projects used to build a model are called the training set and projects used to validate the model are called the validation set. The prediction accuracy of the model was tested on the raw data, using accuracy statistics such as MMRE, Median MRE, Pred(25), the median and mean absolute residuals, and the Companies own estimates (based on an educated guess).

Table 7
Best fitting model to calculate lntoteff

Independent variables	Coefficient	Std error	t	$p > \|t\|$	95% confidence interval
(constant)	2.154	0.260	8.281	0.000	1.634–2.674
lnnewwp	0.435	0.061	7.184	0.000	0.314–0.556
lntoth	0.671	0.160	4.198	0.000	0.352–0.991
devteam	0.239	0.083	2.876	0.005	0.073–0.406

In relation to Companies estimates, their accuracy rate is overall 61.09%, and all underestimates.

The prediction accuracy statistics are presented in Table 8, where we can see that the model's prediction accuracy (MMRE of 99%) was worse than the mean estimate accuracy provided using expert opinion, which was 61.09% (underestimate). Its accuracy was not significantly different from predictions based on the median of the data set (median = 90), confirmed by their MMREs and the Wilcoxon matched-paired signed rank test ($\alpha = 0.05$). The Wilcoxon test was used since the absolute residuals were not normally distributed. These results suggest that the data set is not homogeneous and there are possibly several outliers, which would explain the differences between values obtained for medians and means, for the MREs and absolute residuals. In addition, these results can also be due to non-normally distributed data.

Despite this cost model not presenting good estimation accuracy, its main objective in the context of this paper is to indicate that two of the variables selected by the best fit model are a very close match to the two variables ranked highest in our first survey (S1)—number of Web pages and number of high effort features/functions. We believe this result to be very promising, suggesting that these metrics can be estimated by customers early in the development life cycle and are suitable

Table 8
Prediction accuracy statistics (Mendes and Kitchenham, submitted for publication)

Prediction accuracy	Estimates based on regression model (%)
MMRE	99
Median MRE	70
Pred(25)	9
Mean absolute residual	374.9
Median absolute residual	59.6
MMRE using median effort = 90	194
Estimate accuracy based on expert opinion	61.09

for building Web cost models at, for example, the bidding stage. Note that to obtain the number of high effort functions it is necessary to provide the client with a list of high effort functions and to count those that have been selected. In addition, the development team size must be already known by the Web company.

Finally, in relation to the size metrics and cost drivers that were presented in Section 2, the number of Web pages has been proposed previously as a size metric useful for Web cost estimation. This same metric was identified by our survey S1 as a strong candidate for effort prediction and our model has also selected number of new Web pages. Any of the function-orientated metrics presented in Section 2 were always geared towards measuring functionality using a function points-like approach (e.g. Full-function points, Object points). Our results show that a simpler approach represented simply by a list of features/functions that require high effort to develop from scratch or adapt seems sufficient for early cost estimation, at least based on the data we have gathered. Finally, our model selected the number of people in the development team as a cost driver. It also shows that adding more people to a project does not mean that will finish sooner.

5. Conclusions

This paper investigated metrics that can be used for Web cost estimation at an early stage of a Web development project. One example would be that of estimating effort at the bidding stage, where application requirements are not yet fully known.

Section 2 provided an overview of Web size metrics aimed at Web cost estimation. This overview was then compared to our findings, in Section 4.

Section 3 presented two surveys, a case study, and how the results of these three investigations were used to inform a research project named Tukutuku. The objective of the first survey (S1) was to identify early size metrics and cost factors for Web cost estimation using data collected from 133 Web project price quote forms from companies worldwide. The sample population was that of Web companies that use Web forms to give price quotes on Web development projects. The gathered metrics were ranked and organised into six categories: Web application static metrics, Web application dynamic metrics, Cost drivers, Web project metrics, Web company metrics, and Web interface style metrics. The two most used metrics, according to the ranking, were "total number of Web pages" and "features/functionality to be provided by the application", where the former was used 70% of the time and the latter was used 66% of the time.

A case study and a second survey (S2) were then used to validate the results obtained in S1. Validation in the scope of our work represented identifying size metrics and cost factors important to be used in the Web cost estimation process early in the development life cycle.

The first validation, a case study, consisted of interviewing a mature Web Company. The second validation employed a survey investigation (S2), where the sample population was that of Web companies in New Zealand. Thirty-two Web companies participated in S2. The feedback obtained from Surveys 1 and 2 and the case study was used to prepare Web project data entry forms to gather data on Web projects worldwide. This data gathering initiative was called the Tukutuku benchmarking project. [4] After gathering data on 67 industrial Web projects worldwide we conducted a multivariate regression analysis in order to identify which variables would be selected by the best model fit and to compare these results to those from S1. The two out of three effort predictors selected for the best fit regression model correspond very closely to the two highest ranked size metrics from S1—Web pages and features/functionality.

We believe this result to be very promising, suggesting that these metrics can be estimated by customers early in the development life cycle and are suitable for building Web cost models at, for example, the bidding stage.

Our best fit regression model supports the use of size metrics that can be equally applied to Web hypermedia applications and Web software applications. This result is very encouraging since it suggests that it is possible to obtain a general cost model that suits any Web application.

As part of our future work we are gathering more data on Web projects in order to further the studies presented herein and also to benchmark development productivity and to compare prediction accuracy between cross-company cost models and models built using data from a single company (within-company models).

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[4] http://www.cs.auckland.ac.nz/tukutuku

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